

Assignment 2

Spring 2025 Advanced Machine Learning (BA-64061-001)

Student Name: Sanjar Turaev

Student ID: 811360569

Student e-mail: sturaev@kent.edu

Modifying an Existing Neural Network Model to Improve Performance

In this assignment, I modified an existing neural network model for the IMDB binary classification task to improve its performance with Hyperparameter Tuning. The baseline model consisted of 2 hidden layers with 16 units each, using the ReLU activation function and binary cross-entropy loss. I experimented with various modifications, including:

- Changing the number of hidden layers (1, 2, and 3 layers).
- Changing the number of hidden units (16, 32, and 64 units).
- Using different loss functions (binary cross-entropy vs. MSE).
- Using different activation functions (ReLU vs. tanh).
- Applying regularization techniques (L2 regularization and dropout).

The goal was to improve validation accuracy and reduce validation loss, ensuring better generalization to unseen data.

Below is the summary of all 9 models I experimented.

Model	Configuration	Layers	Units	MSE Loss	ReLU Activation	Binary Cross-Entropy Loss
Model 1	Baseline (2 layers, 16 units)	2	16	No	Yes	Yes
Model 2	1 Hidden Layer	1	16	No	Yes	Yes
Model 3	3 Hidden Layers	3	16	No	Yes	Yes
Model 4	32 Hidden Units	2	32	No	Yes	Yes
Model 5	64 Hidden Units	2	64	No	Yes	Yes
Model 6	MSE Loss Function	2	16	Yes	Yes	No
Model 7	tanh Activation	2	16	No	No (tanh)	Yes
Model 8	L2 Regularization	2	16	No	Yes	Yes (with L2 Regularization)
Model 9	Dropout	2	16	No	Yes	Yes (with Dropout)

Changing the Number of Hidden Layers

- **1 Hidden Layer (Model 2):**
 - **Validation Accuracy:** 88.95%
 - **Validation Loss:** 0.276
 - **Impact:** Reducing the number of layers slightly improved validation accuracy compared to the baseline but increased validation loss, indicating potential underfitting.
- **3 Hidden Layers (Model 3):**
 - **Validation Accuracy:** 88.64%
 - **Validation Loss:** 0.284
 - **Impact:** Adding more layers did not improve validation accuracy and increased validation loss, indicating overfitting.

Changing the Number of Hidden Units

- **32 Hidden Units (Model 4):**
 - **Validation Accuracy:** 88.98%
 - **Validation Loss:** 0.277
 - **Impact:** Increasing the number of units slightly improved validation accuracy but had minimal impact on validation loss.
- **64 Hidden Units (Model 5):**
 - **Validation Accuracy:** 88.76%
 - **Validation Loss:** 0.279
 - **Impact:** Further increasing the number of units did not significantly improve performance and increased the risk of overfitting.

Changing the Loss Function

- **MSE Loss Function (Model 6):**
 - **Validation Accuracy:** 88.67%
 - **Validation Loss:** 0.320
 - **Impact:** Using MSE instead of binary cross-entropy resulted in lower validation accuracy and higher validation loss, indicating that binary cross-entropy is more suitable for this task.

Changing the Activation Function

- **tanh Activation (Model 7):**
 - **Validation Accuracy:** 88.37%
 - **Validation Loss:** 0.290
 - **Impact:** Using tanh instead of ReLU resulted in lower validation accuracy and higher validation loss, indicating that ReLU is more effective for this task.

Applying Regularization Techniques

- **L2 Regularization (Model 8):**
 - **Validation Accuracy:** 88.95%
 - **Validation Loss:** 0.270
 - **Impact:** L2 regularization improved validation accuracy and reduced validation loss compared to the baseline, demonstrating its effectiveness in preventing overfitting.
- **Dropout (Model 9):**
 - **Validation Accuracy:** 89.03%
 - **Validation Loss:** 0.276
 - **Impact:** Dropout also improved validation accuracy and reduced validation loss, showing that it is an effective regularization technique.

Comparison of Model Performance (each model graph are attached on the appendix below)

- Below is a comparison of the models based on **validation accuracy** and **validation loss**:

Model	Configuration	Validation Accuracy	Validation Loss
Model 1	Baseline (2 layers, 16 units)	88.88%	0.275
Model 2	1 Hidden Layer	88.95%	0.276
Model 3	3 Hidden Layers	88.64%	0.284
Model 4	32 Hidden Units	88.98%	0.277
Model 5	64 Hidden Units	88.76%	0.279
Model 6	MSE Loss Function	88.67%	0.320
Model 7	tanh Activation	88.37%	0.290
Model 8	L2 Regularization	88.95%	0.270
Model 9	Dropout	89.03%	0.276

Best Performing Configurations

The **best-performing models** are:

1. **Model 9 (Dropout):**
 - **Validation Accuracy:** 89.03%
 - **Validation Loss:** 0.276
 - **Why:** Dropout effectively prevents overfitting by randomly dropping units during training, improving generalization.
2. **Model 8 (L2 Regularization):**
 - **Validation Accuracy:** 88.95%
 - **Validation Loss:** 0.270
 - **Why:** L2 regularization reduces overfitting by penalizing large weights, leading to better generalization.
3. **Model 4 (32 Hidden Units):**

- **Validation Accuracy:** 88.98%
- **Validation Loss:** 0.277
- **Why:** Increasing the number of units slightly improves the model's capacity to learn complex patterns without overfitting.

Key Insights

1. Regularization Improves Generalization:

- Models with L2 regularization and dropout consistently outperformed the baseline, demonstrating the importance of regularization in preventing overfitting.

2. ReLU is Better Than tanh:

- ReLU activation consistently outperformed tanh, making it the preferred choice for this task.

3. Binary Cross-Entropy is Better Than MSE:

- Binary cross-entropy is more suitable for binary classification tasks, as evidenced by the poor performance of the MSE loss function.

4. Adding More Layers or Units Does Not Always Help:

- Increasing the number of layers or units beyond a certain point did not improve performance and often led to overfitting.

Conclusion

By systematically modifying the baseline model and analyzing the results, I identified that **regularization techniques (L2 and dropout)** and **ReLU activation** significantly improve model performance. The best-performing model used **dropout**, achieving a validation accuracy of **89.03%**. The results highlight the importance of careful hyperparameter tuning and architectural design in achieving optimal outcomes.

Reference

- 1) Lec 4_ Neural Networks Examples.pptx
- 2) Lec 2b_ Neural Networks Building Blocks.pptx

Appendix





